

What is Sequence Alignment?

Sequence alignment is a method used in bioinformatics to arrange two or more biological sequences (DNA, RNA, or Protein) in such a way that **similar regions** can be identified.

Alignment helps us to:

- Find similarity between sequences
- Identify conserved regions
- Understand evolutionary relationships
- Predict function of unknown sequences

Pairwise Sequence Alignment

Pairwise alignment compares **two sequences at a time**.

Example:

Sequence 1: ATGCTAG
Sequence 2: ATGCGAG

It tells us:

- How similar the two sequences are
- Where insertions, deletions, or mismatches occur

There are **two types of pairwise alignment**:

1. Global Alignment

Global alignment aligns **two sequences from start to end**, over their entire length.

- Best when sequences are **similar and of equal length**
- Tries to match **every character**

Key Points:

- Full-length alignment
- Penalizes gaps heavily
- Useful for comparing closely related sequences

Example:

Seq 1: ATGCTAG
Seq 2: ATGC-AG

Common Algorithm:

- Needleman–Wunsch algorithm

When to Use:

- Comparing genes from the same species
- Comparing similar proteins

2. Local Alignment

Local alignment finds the **best matching region** between two sequences.

- Does **not** align the entire sequence
- Focuses only on highly similar parts

Key Points:

- Partial alignment
- Ignores weakly matching regions
- Very useful when sequences are **different in length**

Example:

```
Seq 1: ATGCTAGGCTA
Seq 2:   CTAGGC
```

Common Algorithm:

- **Smith–Waterman algorithm**

When to Use:

- Finding conserved domains
- BLAST uses local alignment

3. Multiple Sequence Alignment (MSA)

Multiple sequence alignment aligns **three or more sequences together** to find conserved regions among all sequences.

Purpose:

- Identify conserved motifs
- Study evolutionary relationships
- Build phylogenetic trees

Example:

```
Seq1: ATGC-TAG
Seq2: ATGCGTAG
Seq3: ATG--TAG
```

Common Tools:

- ClustalW
- MUSCLE
- MAFFT

Similarity Score

A **similarity score** is a numerical value that tells **how similar two sequences are** after alignment.

How is it Calculated?

- **Match** → positive score
- **Mismatch** → negative score
- **Gap** → penalty

Example:

Alignment Element	Score
Match	+1
Mismatch	-1
Gap	-2

Higher score = more similarity
Lower score = less similarity

Why is Alignment Important in Bioanalysis?

Alignment helps in:

- Identifying homologous genes
- Predicting protein function
- Detecting mutations
- Understanding evolutionary biology

Summary Table

Feature	Global Alignment	Local Alignment
Aligns full sequence	Yes	No
Best for	Similar sequences	Partially similar sequences
Algorithm	Needleman–Wunsch	Smith–Waterman
Used in BLAST	No	Yes

Conclusion

- **Global alignment** compares sequences end-to-end
- **Local alignment** finds best matching regions
- **Multiple sequence alignment** compares many sequences

- **Similarity score** measures how close sequences are

These concepts are **core skills** for a **Bioanalyst / Bioinformatics student**.

What is Global Alignment?

Global alignment aligns two sequences from start to end (full length comparison).

It is best used when:

- Sequences are almost the same length
- Sequences are highly similar

Real-life example

Imagine two sentences:

```
HELLO WORLD  
HELLO WORD
```

Global alignment compares the entire sentence, even if there is a mismatch at the end.

Bio example

```
Seq1: ATGCT  
Seq2: ATGCA
```

The entire sequence **ATGCT vs ATGCA** will be compared.

What is Local Alignment?

Local alignment finds only the best matching region between two sequences.

It is best used when:

- Sequences have different lengths
- Only a portion of the sequences is similar

Real-life example

```
Sentence 1: I LOVE BIOINFORMATICS  
Sentence 2: BIOINFORMATICS IS FUN
```

Local alignment will detect:

"BIOINFORMATICS" as the common matching region

Bio example

```
Seq1: ATGCTAGCTA
```

Seq2: GCTAGC

Only **GCTAGC** will be aligned.

What is Similarity Score?

After alignment, we get a numerical score.

This score tells us how good the alignment is.

The score depends on:

- Match (+ score)
- Mismatch (− penalty)
- Gap (− penalty)

Example scoring scheme

```
Match = +1  
Mismatch = -1  
Gap = -2
```

Higher score = better similarity

How to Perform Alignment in Biopython?

In Biopython, we use:

```
Bio.Align.PairwiseAligner
```

Step 1: Import the Aligner

```
from Bio.Align import PairwiseAligner
```

Step 2: Create an Aligner Object

```
aligner = PairwiseAligner()
```

Step 3: Global Alignment (Default Mode)

```
seq1 = "ATGCT"  
seq2 = "ATGCA"  
  
alignments = aligner.align(seq1, seq2)  
  
print(alignments[0])
```

This performs global alignment.
`alignments[0]` gives the best alignment.

Step 4: Local Alignment

```

aligner.mode = "local"

seq1 = "ATGCTAGCTA"
seq2 = "GCTAGC"

alignments = aligner.align(seq1, seq2)

print(alignments[0])

```

This shows only the best matching region.

Step 5: Check Similarity Score

```

alignment = alignments[0]
print("Score:", alignment.score)

```

Setting Match, Mismatch, and Gap Penalties

```

aligner.match_score = 1
aligner.mismatch_score = -1
aligner.open_gap_score = -2
aligner.extend_gap_score = -0.5

```

Summary (Exam + Interview Ready)

Concept	Meaning
Global Alignment	Full sequence vs full sequence
Local Alignment	Only the best matching region
Similarity Score	Numerical value representing alignment quality
Biopython Tool	PairwiseAligner

What are Biological Databases?

Biological databases are online repositories that store biological information such as:

- DNA sequences
- RNA sequences
- Protein sequences
- Annotations
- Structural data

They are essential for **bioanalysis**, research, and computational biology.

NCBI (National Center for Biotechnology Information)

NCBI is a public biological database maintained by the **National Institutes of Health (NIH), USA**.

It provides access to:

- DNA & RNA sequences
- Protein sequences
- Literature (PubMed)
- Analysis tools (BLAST)

Why NCBI is Important?

- Central hub for biological data
- Most bioinformatics tools depend on NCBI
- Free and globally used

What is GenBank?

GenBank is a nucleotide sequence database maintained by NCBI.

It stores:

- DNA sequences
- RNA sequences
- Annotations (gene, CDS, exon, intron)

File Format:

- .gb or .gbk

Contains:

- Sequence
- Features
- Source organism
- References

Used for: Gene annotation, genome analysis

RefSeq (Reference Sequence Database)

RefSeq is a **curated and non-redundant** database provided by NCBI.

Key Features:

- High-quality reference sequences
- Fewer errors than GenBank
- One reference per gene

Used for:

- Comparative genomics
- Accurate gene analysis

Difference from GenBank:

- GenBank → submitted data
- RefSeq → reviewed data

UniProt (Universal Protein Resource)

UniProt is a protein sequence and functional information database.

Stores:

- Protein sequences
- Protein function
- Domains
- Post-translational modifications

Two Main Sections:

- **UniProtKB/Swiss-Prot** → manually reviewed
- **UniProtKB/TrEMBL** → automatically annotated

Used for: Protein analysis & functional prediction

PDB (Protein Data Bank)

PDB stores **3D structural data** of biological molecules.

Contains:

- Protein structures
- DNA/RNA structures
- Complexes

Structure Types:

- X-ray crystallography
- NMR
- Cryo-EM

Used for:

- Structural bioinformatics
- Drug design

Why Database Knowledge is Important for Bioanalysts?

A bioanalyst must:

- Find reliable biological data
- Compare sequences across species

- Analyze structure and function
- Validate experimental results

Biopython for Database Handling

Biopython allows direct access to biological databases.

Module used:

```
from Bio import Entrez
```

What You Can Do:

- Search databases
- Download sequences
- Fetch by accession ID

Accession ID Fetching

An **Accession ID** is a unique identifier given to each sequence in a database.

Example:

- NM_000546 (RefSeq)
- NC_045512 (GenBank)
- P69905 (UniProt)

Used to retrieve exact sequences without confusion.

FASTA and GenBank Parsing

Parsing means reading and extracting information from sequence files.

File Formats:

- FASTA (.fasta)
- GenBank (.gb)

Biopython Module:

```
from Bio import SeqIO
```

What Can Be Parsed?

- Sequence length
- Features
- Annotations
- Organism name

Why Use Biopython?

Biopython helps to:

- Automate sequence analysis
- Handle large datasets
- Reduce manual errors
- Perform professional bioanalysis

Summary Table

Database	Stores	Used For
NCBI	All biological data	Central resource
GenBank	DNA/RNA sequences	Gene annotation
RefSeq	Curated sequences	Reference analysis
UniProt	Protein data	Functional study
PDB	3D structures	Structural biology

What is Entrez in Biopython?

Entrez is a programming interface provided by **NCBI** that allows you to:

- Search biological databases
- Download sequences
- Fetch data using Accession IDs

In simple words:

Entrez = Bridge between Python and NCBI databases

Why Entrez is Important for Bioanalysts?

Using Entrez, you can:

- Automatically download sequences
- Avoid manual copy-paste
- Work with large datasets
- Perform real bioinformatics analysis

Step 1: Import Entrez Module

```
from Bio import Entrez
```

Step 2: Set Email (VERY IMPORTANT)

NCBI requires your email for tracking usage.

```
Entrez.email = "your_email@example.com"
```

Without this, NCBI may block your requests.

Step 3: Search a Database (Entrez.esearch)

Example: Search gene sequences in NCBI Nucleotide database

```
handle = Entrez.esearch(  
    db="nucleotide",  
    term="BRCA1[Gene] AND Homo sapiens[Organism]"  
)  
result = Entrez.read(handle)  
print(result)
```

Output contains:

- Count of results
- List of IDs (GI / accession numbers)

Step 4: Fetch Sequence Using Accession ID (Entrez.efetch)

Example: Fetch DNA sequence in FASTA format

```
handle = Entrez.efetch(  
    db="nucleotide",  
    id="NM_007294",  
    rettype="fasta",  
    retmode="text"  
)  
sequence = handle.read()  
print(sequence)
```

This downloads the sequence directly from NCBI.

Step 5: Save Downloaded Sequence to a File

```
with open("sequence.fasta", "w") as file:  
    file.write(sequence)
```

Fetch GenBank File Using Accession ID

```
handle = Entrez.efetch(  
    db="nucleotide",  
    id="NM_007294",  
    rettype="gb",  
    retmode="text"  
)  
genbank_data = handle.read()  
print(genbank_data)
```

This file contains:

- Sequence
- Features
- Annotations

- References

FASTA & GenBank Parsing Using Biopython

Parse FASTA File

```
from Bio import SeqIO

for record in SeqIO.parse("sequence.fasta", "fasta"):
    print("ID:", record.id)
    print("Length:", len(record.seq))
    print("Sequence:", record.seq)
```

Parse GenBank File

```
for record in SeqIO.parse("sequence.gb", "genbank"):
    print("Organism:", record.annotations["organism"])
    print("Sequence length:", len(record.seq))
```

Extract Features from GenBank File

```
for feature in record.features:
    print("Feature type:", feature.type)
```

Extract CDS (Coding Sequences)

```
for feature in record.features:
    if feature.type == "CDS":
        print("Gene:", feature.qualifiers.get("gene"))
        print("Product:", feature.qualifiers.get("product"))
```

Convert GenBank to FASTA

```
SeqIO.convert("sequence.gb", "genbank", "output.fasta", "fasta")
```

Real-Life Bioanalyst Workflow Example

- Get Accession ID from paper or BLAST
- Fetch sequence using Entrez
- Save FASTA / GenBank
- Parse file using SeqIO
- Analyze features and sequences

Summary (Exam-Friendly)

- **Entrez** allows programmatic access to NCBI
- `esearch()` → search database
- `efetch()` → download sequence
- **Accession ID** uniquely identifies sequences
- **SeqIO** parses FASTA and GenBank files
- Feature extraction is key for gene analysis